

that required to activate the relay. In this condition, when you press switch S2, the relay does not energise to provide mains power to the device. Thus the circuit offers some sort of under-voltage protection.

The combination of a step-down transformer, a rectifier and a filter capacitor provides the required voltage for activation of relay RL1. Push-to-off switch S1 is used to break the mains

power supply from the load (switch off the load). Push-to-on switch S2 is used to connect the mains power supply to the load (switch-on the load). A freewheeling diode across the relay coil prevents the back e.m.f., whereas LED1 gives an indication of relay energisation.

Working of the circuit is simple. When you press switch S2 momentarily, current flows through the relay coil

to energise it. As a result, normally-open (N/O) contact of the relay closes to connect the load to the power. The relay de-energises if the power fails or you press switch S1 momentarily to cut the current path of the relay.

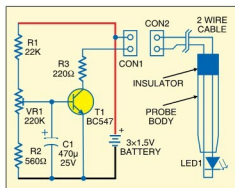
Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Mount switches S1 and S2 on the front panel and the relay at the rear side to connect the load. ●

WHITE LED LIGHT PROBE FOR INSPECTION

■ M.K. CHANDRA MOULEESWARAN

This circuit is useful for inspecting narrow spaces like the inside of the CPUs, monitors, PCB modules and other electronic devices. The light source is a pencil-tube tube with ultra-bright white LED at the tip.

The circuit is built around transistor T1 to control the brightness of LED1. The forward-bias voltage of the transistor is set using preset VR1. The centre pin of preset VR1 is connected to the base of transistor T1. Resistors R1 and R2 and preset VR1 form a voltage-



divider circuit to control the brightness of LED1. Varying preset VR1 varies the forward bias voltage and thus the collector current of transistor T1. This way, the brightness of white LED1 is

controlled. Resistor R3 in the collector circuit of T1 limits the current through LED1.

The ultra-bright LED requires up to 20 mA for full brightness. So there is no need for a separate on/off switch. Disconnecting CON1 from CON2 is enough when not in use.

Assemble the circuit on a small piece of PCB and solder a small two-pin female connector (CON1) on its right side for connecting the probe. For the probe, use the cabinet of a plastic pen. Fit the white LED at one end and a small two-pin male connector (CON2) at the other end of the pen. ●



FOUR-STAGE FM TRANSMITTER

■ PRADEEP G.

This FM transmitter circuit uses four radio frequency stages: a VHF oscillator built around transistor BF494 (T1), a preamplifier built around transistor BF200 (T2), a driver built around transistor 2N2219 (T3) and a power amplifier built around transistor 2N3866 (T4). A condenser microphone is connected at the input of the oscillator.

Working of the circuit is simple. When you speak near the microphone, frequency-modulated signals are obtained at the collector of oscillator transistor T1. The FM signals are amplified by the VHF preamplifier and the pre-driver stage. You can also use transistor 2N5109 in place of 2N2219. The preamplifier is a tuned class-A RF amplifier and the driver is a class-C amplifier. Signals are finally fed to the class-C RF power amplifier, which de-

livers RF power to a 50-ohm horizontal dipole or ground plane antenna.

Use a heat-sink with transistor 2N3866 for heat dissipation. Carefully adjust trimmer VC1 connected across L1 to generate frequency within 88-108 MHz. Also adjust trimmers VC2 through VC7 to get maximum output at maximum range.

Regulator IC 78C09 provides stable

